

# Year Two Lecture Notes

BSc Computer Science - Durham University

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## Part I

# Artificial Intelligence

## 1 Module Introduction

Artificial Intelligence has the module code COMP2261, and is led by Dr Yang Long.

### 1.1 Assessment

This module is assessed through three pieces of summative coursework, these can be seen in Table 1. **Final dates to be confirmed.**

| Title            | Opening    | Due        | Feedback   | Professor            | Weight |
|------------------|------------|------------|------------|----------------------|--------|
| Machine Learning | 10/10/2026 | 16/12/2026 | 22/01/2026 | Dr Yang Long         | 50%    |
| AI Search        | 16/01/2027 | 19/02/2027 | 19/03/2026 | Prof. Iain Stewart   | 25%    |
| Bias in AI       | 06/02/2027 | 26/03/2027 | 07/05/2027 | Dr. Shauna Concannon | 25%    |

Table 1: Assessments for Artificial Intelligence.

### 1.2 Timetable

This module is taught through two one-hour lectures and one two-hour computer class each week as seen below in Table 2. **Final times and locations to be confirmed.**

| Day     | Time          | Class Type     | Location | Lecturer     |
|---------|---------------|----------------|----------|--------------|
| Monday  | 14:00 - 16:00 | Computer Class | TLC025   | n/a          |
| Tuesday | 16:00 - 17:00 | Lecture        | W103     | Dr Yang Long |
| Friday  | 15:00 - 16:00 | Lecture        | W103     | Dr Yang Long |

Table 2: Michaelmas Term Timetable for Artificial Intelligence.

## 2 Machine Learning

## 3 Artificial Intelligence Search

## 4 Bias in AI

## Part II

# Data Science

## 5 Module Introduction

Data Science has the module code COMP2271 and is led by Professor Thomas Erlebach.

### 5.1 Assessment

This module is assessed through two pieces of summative coursework and two final exams, these can be seen in Table 3. **Final dates to be confirmed.**

| Title                     | Opening    | Due        | Feedback   | Professor            | Weight |
|---------------------------|------------|------------|------------|----------------------|--------|
| Data Cleaning & Analytics | Week 11    | n/a        | 13/02/2027 | Dr Igor Razgon       | 25%    |
| Image Processing          | 16/01/2027 | 26/02/2027 | 26/03/2027 | Dr George Koulrieris | 25%    |
| Probability and Graphics  | May Exam   | n/a        | n/a        | Razgon & Koulrieris  | 50%    |

Table 3: Assessments for Data Science.

### 5.2 Timetable

This module is taught through two one-hour lectures and one two-hour computer class each week as seen below in Table 4.

| Day    | Time          | Class Type     | Location | Lecturer       |
|--------|---------------|----------------|----------|----------------|
| Monday | 12:00 - 13:00 | Lecture        | W103     | Dr Igor Razgon |
| Friday | 10:00 - 11:00 | Lecture        | W103     | Dr Igor Razgon |
| Friday | 16:00 - 18:00 | Computer Class | CC-0007  | n/a            |

Table 4: Michaelmas Term Timetable for Data Science.

## 6 Probability and Statistics

This section is taught by Dr Igor Razgon.

## 7 Data Cleaning and Analytics

This section is taught by Dr Igor Razgon.

## 8 Computer Graphics

This section is taught by Dr George Alex Koulrieris.

## 9 Image Processing

This section is taught by Dr George Alex Koulrieris.

## Part III

# Networks and Systems

## 10 Module Introduction

Networks and Systems has the module code COMP2211 and is led by Dr Wanqing Tu.

### 10.1 Assessment

This module is assessed through two pieces of summative coursework and two final exams, these can be seen in Table 5. **Final dates to be confirmed.**

| Title                  | Opening    | Due        | Feedback   | Professor               | Weight |
|------------------------|------------|------------|------------|-------------------------|--------|
| Cyber Security         | 10/10/2026 | 06/11/2026 | 04/12/2026 | Dr Charles Morisset     | 25%    |
| Networks               | 31/10/2026 | 08/01/2027 | 05/02/2027 | Dr Wanqing Tu           | 25%    |
| Dist. Sys. & Databases | May Exam   | n/a        | n/a        | <b>To be confirmed.</b> | 50%    |

Table 5: Assessments for Networks and Systems.

### 10.2 Timetable

This module is taught through one two-hour lecture and one two-hour computer class each week as seen below in Table 6. **Final times and locations to be confirmed.**

| Day      | Time          | Class Type     | Location | Lecturer            |
|----------|---------------|----------------|----------|---------------------|
| Thursday | 09:00 - 11:00 | Lecture        | CLC202   | Dr Carlton Shepherd |
| Thursday | 16:00 - 18:00 | Computer Class | CC-0007  | Dr Carlton Shepherd |

Table 6: Michaelmas Term Timetable for Networks and Systems.

## 11 Cyber Security

## 12 Networks

## 13 Distributed Systems

## 14 Databases

## Part IV

# Programming Paradigms

## 15 Module Introduction

Programming Paradigms has the module code COMP2221 and is led by Dr Stuart James.

### 15.1 Assessment

This module is assessed through one piece of summative coursework and two final exams, these can be seen in Table 7. **Final dates to be confirmed.**

| Title                        | Opening    | Due        | Feedback   | Professor       | Weight |
|------------------------------|------------|------------|------------|-----------------|--------|
| Systems Programming          | 31/10/2026 | 11/12/2026 | 08/01/2027 | Dr Stuart James | 50%    |
| Object Oriented & Functional | May Exam   | n/a        | n/a        | Ivkovic & Moses | 50%    |

Table 7: Assessments for Programming Paradigms.

### 15.2 Timetable

This module is taught through two one-hour lectures and one two-hour computer class each week as seen below in Table 8.

| Day    | Time          | Class Type     | Location | Lecturer        |
|--------|---------------|----------------|----------|-----------------|
| Monday | 11:00 - 12:00 | Lecture        | CLC202   | Dr Stuart James |
| Monday | 16:00 - 18:00 | Computer Class | CB-LG001 | n/a             |
| Friday | 14:00 - 15:00 | Lecture        | CLC202   | Dr Stuart James |

Table 8: Michaelmas Term Timetable for Programming Paradigms.

## 16 Systems Programming

This section is taught by Dr Stuart James.

## 17 Object Oriented Programming

This section is taught by Dr Mladen Ivkovic.

## 18 Functional Programming

This section is taught by Dr Billy Moses Jr..



## Part V

# Software Engineering

### 19 Module Introduction

Software Engineering has the module code COMP2281 and is led by Dr Craig Stewart.

#### 19.1 Assessment

This module is assessed through a group project, the deadlines for this can be seen in Table 9. **Final dates to be confirmed.**

| Title                        | Opening    | Due        | Feedback   | Professor        | Weight |
|------------------------------|------------|------------|------------|------------------|--------|
| Peer Evaluation 1            | 17/10/2026 | 20/11/2026 | 22/01/2027 | Dr Craig Stewart | 1.67%  |
| Requirement Documentation    | 17/10/2026 | 20/11/2026 | 22/01/2027 | Dr Craig Stewart | 15%    |
| Peer Evaluation 2            | 17/10/2026 | 12/02/2027 | 07/05/2027 | Dr Craig Stewart | 1.67%  |
| Technical Report             | 17/10/2026 | 12/03/2027 | 07/05/2027 | Dr Craig Stewart | 30%    |
| Product Presentation         | 17/10/2026 | 20/03/2027 | 07/05/2027 | Dr Craig Stewart | 15%    |
| Peer Evaluation 3            | 17/10/2026 | 24/03/2027 | 07/05/2027 | Dr Craig Stewart | 1.67%  |
| Reflective Experience Report | 17/10/2026 | 30/04/2027 | n/a        | Dr Craig Stewart | 20%    |
| Project Handover             | 17/10/2026 | 30/04/2027 | n/a        | Dr Craig Stewart | 10%    |
| Git Commit                   | 17/10/2026 | 30/04/2027 | n/a        | Dr Craig Stewart | 5%     |

Table 9: Assessments for Software Engineering.

#### 19.2 Timetable

This module is taught through two one-hour lectures and one two-hour computer class each week as seen below in Table 10. **Final times and locations to be confirmed.**

| Day       | Time          | Class Type | Location | Lecturer           |
|-----------|---------------|------------|----------|--------------------|
| Tuesday   | 09:00 - 10:00 | Lecture    | TLC033   | Dr Nelly Bencomo   |
| Wednesday | 11:00 - 12:00 | Lecture    | TLC033   | Dr Nelly Bencomo   |
| Thursday  | 11:00 - 13:00 | Practical  | CC-0007  | Dr Gagangeet Aujla |

Table 10: Michaelmas Term Timetable for Software Engineering.

## **20 Planning and Design**

### **20.1 Requirement Engineering**

### **20.2 Modelling and Analysis**

### **20.3 Software Design**

## **21 Development and Testing**

### **21.1 Software Development Methodologies**

### **21.2 Software Testing**

## **22 Development Considerations**

### **22.1 Software Quality Management**

### **22.2 Human Computer Interaction**

### **22.3 Legal, Social, Ethical, and Professional Issues**

## **23 Group Project**

### **23.1 Introduction**

### **23.2 Project Management**

## Part VI

# Theory of Computation

## 24 Module Introduction

Theory of Computation has the module code COMP2181 and is led by Dr Barnaby Martin.

### 24.1 Assessment

This module is assessed through two summative benchtests, and one final exams, these can be seen in Table 11.

| Title       | Opening  | Due | Feedback   | Professor                   | Weight |
|-------------|----------|-----|------------|-----------------------------|--------|
| Benchtest 1 | Week 11  | n/a | 09/02/2027 | Dr Dantchev and Dr Mertzios | 17%    |
| Benchtest 2 | Week 20  | n/a | 07/05/2027 | Dr Dantchev and Dr Mertzios | 17%    |
| Final Exam  | May Exam | n/a | n/a        | Dr Barnaby Martin           | 66%    |

Table 11: Assessments for Theory of Computation.

### 24.2 Timetable

This module is taught through two one-hour lectures and one two-hour computer class each week as seen below in Table 12.

| Day       | Time          | Class Type    | Location | Lecturer          |
|-----------|---------------|---------------|----------|-------------------|
| Tuesday   | 10:00 - 11:00 | Lecture       | CLC202   | Dr Barnaby Martin |
| Wednesday | 09:00 - 11:00 | Problem Class | TLC116   | n/a               |
| Thursday  | 14:00 - 15:00 | Lecture       | CLC202   | Dr Sukanya Pandey |

Table 12: Michaelmas Term Timetable for Theory of Computation.

## 25 Models of Computation

This section is taught by Dr Barnaby Martin.

### 25.1 Finite-State Automata and Regular Languages

### 25.2 Büchi Automata and Omega-Regular Languages

### 25.3 Linear-Time Logic and Translating Into Büchi Automata

## 26 Algorithms and Complexity

### 26.1 Asymptotic Complexity and Notation

### 26.2 Greedy Algorithms

### 26.3 Divide and Conquer Algorithms

### 26.4 Dynamic Programming Algorithms

### 26.5 Machinings in Graphs

## Part VII

# Appendix

## 27 Artificial Intelligence

### 27.1 Module Professors

**Module Lead:** Doctor Yang Long

- Focus: Scalable AI solutions for large-scale healthcare applications.
- Email: yang.long@durham.ac.uk
- Office Hours: **To be confirmed.**

**Lecturer:** Professor Iain Stewart

- Focus: Computational complexity, graph/group theory, and theoretical aspects artificial intelligence.
- Email: i.a.stewart@durham.ac.uk
- Office Hours: **To be confirmed.**

**Lecturer:** Doctor Shauna Concannon

- Focus: Computational linguistics, social interaction, and human-computer interaction.
- Email: shauna.j.concannon@durham.ac.uk
- Office Hours: **To be confirmed.**

### 27.2 References

### 27.3 List of Figures

## 28 Data Science

### 28.1 Module Professors

**Module Lead:** Professor Thoas Erlebach

- Focus: Algorithms and networks.
- Email: thomas.erlebach@durham.ac.uk
- Office Hours: n/a.

**Lecturer:** Doctor Igor Razgon

- Focus: Algorithms and complexity, graph theory, and discrete mathematics.
- Email: igor.razgon@durham.ac.uk
- Office Hours: Friday 14:00-15:00.

**Lecturer:** Doctor George Alex Koulieris

- Focus: Image processing and computer graphics.
- Email: georgios.a.koulieris@durham.ac.uk
- Office Hours: Please email to make an appointment.

## 28.2 References

## 28.3 List of Figures

# 29 Networks and Systems

## 29.1 Module Professors

**Module Lead:** Doctor Wanqing Tu

- Focus: Computer networks and wireless communications.
- Email: wanqing.tu@durham.ac.uk
- Office Hours: **To be confirmed.**

**Lecturer:** Doctor Charles Morisset

- Focus: Explainable security, access control, and the security and privacy of smart infrastructure.
- Email: charles.morisset@durham.ac.uk
- Office Hours: **To be confirmed.**

**Lecturer:** Doctor Carlton Shepherd

- Focus: Cyber security.
- Email: carlton.g.shepherd@durham.ac.uk
- Office Hours: **To be confirmed.**

## 29.2 References

## 29.3 List of Figures

# 30 Programming Paradigms

## 30.1 Module Professors

**Module Lead:** Doctor Stuart James

- Focus: Artificial intelligence, computer vision, human computer interaction, and digital humanities.
- Email: stuart.a.james@durham.ac.uk
- Office Hours: Email to make an appointment.

**Lecturer:** Doctor Billy Moses Jr.

- Focus: Algorithms and biologically inspired computing.
- Email: [william.k.moses-jr@durham.ac.uk](mailto:william.k.moses-jr@durham.ac.uk)
- Office Hours: Email to make an appointment.

**Lecturer:** Doctor Mladen Ivkovic

- Focus: Performance prediction and optimisation.
- Email: [mladen.ivkovic@durham.ac.uk](mailto:mladen.ivkovic@durham.ac.uk)
- Office Hours: Email to make an appointment.

## 30.2 References

## 30.3 List of Figures

# 31 Software Engineering

## 31.1 Module Professors

**Module Lead:** Doctor Craig Stewart

- Focus: Adaptive systems, HCI, and user modelling.
- Email: [craig.d.stewart@durham.ac.uk](mailto:craig.d.stewart@durham.ac.uk)
- Office Hours: **To be confirmed.**

**Lecturer:** Doctor Nelly Bencomo

- Focus: Software engineering for autonomous, self-adaptive, AI and ML systems.
- Email: [nelly.bencomo@durham.ac.uk](mailto:nelly.bencomo@durham.ac.uk)
- Office Hours: **To be confirmed.**

**Lecturer:** Doctor Gagangeet Aujla

- Focus: Data-driven and data-intensive cyber-physical systems and digital twins.
- Email: [gagangeet.s.aujla@durham.ac.uk](mailto:gagangeet.s.aujla@durham.ac.uk)
- Office Hours: **To be confirmed.**

## 31.2 References

## 31.3 List of Figures

# 32 Theory of Computation

## 32.1 Module Professors

**Module Lead:** Doctor Barnaby Martin

- Focus: Complexity theory, finite model theory, and the links between logic and complexity.

- Email: barnaby.d.martin@durham.ac.uk
- Office Hours: Flexible, contact to arrange.

**Lecturer:** Doctor Sukanya Pandey

- Focus: Not specified.
- Email: sukanya.pandey@durham.ac.uk
- Office Hours: Flexible, contact to arrange.

**Lecturer:** Doctor Stefan Dantchev

- Focus: Combinatorics.
- Email: s.s.dantchev@durham.ac.uk
- Office Hours: **To be confirmed.**

**Lecturer:** Doctor George Mertzios

- Focus: Algorithms and networks.
- Email: george.mertzios@durham.ac.uk
- Office Hours: **To be confirmed.**

## 32.2 References

## 32.3 List of Figures

## 32.4 Recommended Reading

Cormen, T.H., Leiserson, C.E., Rivest, R.L., and Stein, C. (2009) *Introduction to Algorithms - Third Edition*. MIT Press.